Higher National Diploma in Information Technology
First Year, Second Semester Examination – 2020
HNDIT1209 Object Oriented Programming.

Instructions for Candidates:
Answer only five questions

No. of questions: 6
No. of pages : 4
Time : 3 hours

Question 01

I. What is meant by high-level programming language? (02 marks)

II. "Java is a platform independent language" justify the above statement. (02 marks)

III. Give an example for the following categories of literals. (02 marks)
   a. char literal
   b. string literal.

IV. Identify the following literal types (02 marks)
   a. 0.23f
   b. 0XABC

V. Write the output of following java code segments (02 marks)

(2x6=12 marks)

a.
```
System.out.print("nn\nn");
```

b.
```
int V1=12,V2=12;
System.out.print(V1%V2);
```

c.
```
System.out.print("bb\bb");
```

d.
```
int a , b;
a= 10;
b= 10;System.out.println((a == 1) ? 20: 30);
```

e.
```
int aa=0;
System.out.print(a++);
System.out.print(++a);
```

f.
```
System.out.print(2/2+2-2*2);
```

(Total 20 marks)

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Question 02

I. What is meant by control flow? (02 marks)
II. What is meant by repetition structure? (02 marks)
III. Draw the flow chart of "for" loop. (02 marks)
IV. What is the use of "break" statement in java? (02 marks)
V. What would be the output of the code segments given below? (6+6 =12 marks)

a.
```
for(int a=0;a<7;a++)
    {
    for(int b=0;b<a;b++)
        {
        System.out.print(b);
        }
    System.out.println("");
    }
```

b.
```
for(int a=7;a>0;a--)
{
for(int b=7;b>a;b--)
    {
    System.out.print(b);
    }
System.out.println("");
}
```

(Total 20 marks)

Question 03

I. Explain the differences between class and object in object-oriented programming. (4 marks)

II. What is constructor? (2 marks)

III. What is meant by access modifier in java programming ? (2 marks)

IV. Write the java coding for following requirements

a. Create a class named **"WaterBill"** (2 marks)

b. Private fields "**PreviousReading, currentReading, consumption** and **charge**" to store the previous meter reading, current meter reading, consumption and charge. (2 marks)

c. Public function **"calConsumption()"** to calculate the water consumption. Use formula consumption = currentReading - PreviousReading (2 marks)

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d. Public function **"calWaterCharge()"** to calculate the charge using the given table below.

(6 marks)

Number of units	Usage charge per unit	Service charge
0 - 5	12.00	50
6 - 10	16.00	65
More than 11	20.00	70

(Total 20 marks)

Question 04

I. What is meant by inheritance in object-oriented programming? (02 marks)
II. Mention two types of inheritance. (02 marks)
III. Mention two advantages of inheritance. (02 marks)
IV. Explain the purpose of "super" key word (02 marks)

V. UniTel is a telecommunication company. The monthly rental of package named **LiteUser** is Rs.500.00, and 1000 minutes of outgoing calls are free of charge for any network. Outgoing calls other than 1000 minutes are charged as Rs 0.20 per minute.

a. Write the java code for the **LiteUser** class including two private fields named **charge** and **minutes.**

(03 marks)

b. Write the Java code for method named **PrintData()** to print the number of minutes and charges, which is can't override in child classes.

(03 marks)

c. Write the java code for method named **calculateBill()** to process the calculation, which can be override in child classes.

(04 marks)

d. Write the java code for child named **HeavyUser** inherited from **LiteUser** class.

(02 marks)

(Total 20 marks)

Question 05

I. What is meant by abstraction?

(2 marks)

II. What is the advantage of abstraction?

(2 marks)

III. Compare and contrast abstract class and interface class.

(4 marks)

IV. Write the suitable java code for the following statements.

a. Abstract class named **Item** with attribute **itemPrice** , which is float type.

(2 marks)

b. **Item** class has a public method **print()** to print the price.

(2 marks)

c. **Item** class has an abstract method **reselPrice()**

(2 marks)

d. Child class named **DistributedItem** inherited from abstract class **Item**

(2 marks)

e. Implement the abstract method **reselPrice()** to increase the **itemPrice** by 5%

(4 marks)

(Total 20 marks)

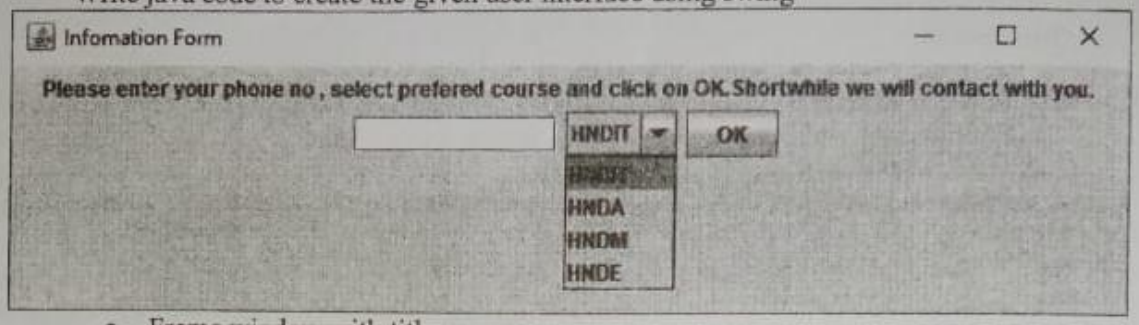
Question 06

I. What is meant by swing in java (2 marks)

II. Mention two features in swing. (2 marks)

III. Mention two advantages of using swing in contrast to AWT (2 marks)

Write java code to create the given user interface using swing



a. Frame window with title (2 marks)

b. Create and add the label (3 marks)

c. Create and add the button with action listener (4 marks)

d. Create and add the combo box with data (5 marks)

(Total 20 marks)

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